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Appendix

Additional details and results from the different sections are included below.

A Additional details on Methods and the Experimental Setup

To understand systematic differences between ViT and CNNs, we use a representative set of different models of each type, guided by the performance results in [14]. Specifically, for ViTs, we look at ViT-B/32, ViT-L/16 and ViT-H/14, where the smallest model (ViT-B/32) shows limited improvements when pretraining on JFT-300M [40] vs. the ImageNet Large Scale Visual Recognition Challenge 2012 dataset [12, 37], while the largest, ViT-H/14, achieves state of the art when pretrained on JFT-300M [40]. ViT-L/16 is close to the performance ViT-H/14 [14]. For CNNs, we look at ResNet50x1 which also shows saturating performance when pretraining on JFT-300M, and also ResNet152x2, which in contrast shows large performance gains with increased pretraining dataset size. As in Dosovitskiy et al. [14], these ResNets follow some of the implementation changes first proposed in BiT [16].

In addition to the standard Vision Transformers trained with a classification token (CLS), we also trained ViTs with global average pooling (GAP). In these, there is no classification token – instead, the representations of tokens in the last layer of the transformer are averaged and directly used to predict the logits. The GAP model is trained with the same hyperparameters as the CLS one, except for the initial learning rate that is set to a lower value of 0.0003.

For analyses of internal model representations, we observed no meaningful difference between representations of images drawn from ImageNet and images drawn from JFT-300M. Figures 1, 2, 9, and 10 use images from the JFT-300M dataset that were not seen during training, while Figures 6, 3, 4, 5 7, 8, and 12 use images from the ImageNet 2012 validation set. Figures 11 and 13 involve 10-shot probes trained on the ImageNet 2012 training set, tuned hyperparameters on a heldout portion of the training set, and evaluated on the validation set.

Additional details on CKA implementation To compute CKA similarity scores, we use minibatch CKA, introduced in [29]. Specifically, we use batch sizes of 1024 and we sample a total of 10240 examples without replacement. We repeat this 20 times and take the average. Experiments varying the exact batch size (down to 128), and total number of examples used for CKA (down to 2560 total examples, repeated 10 times), had no noticeable effect on the results.

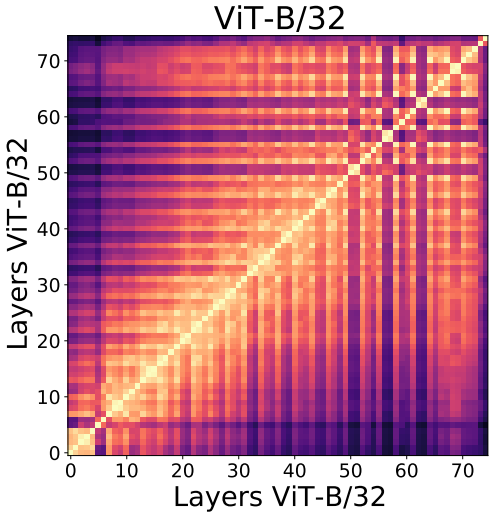
Additional details on linear probes. We train linear probes as regularized least-squares regression, following Dosovitskiy et al. [14]. We map the representations training images to $\{-1, 1\}^N$ target vectors, where N is the number of classes. The solution can be recovered efficiently in closed form.

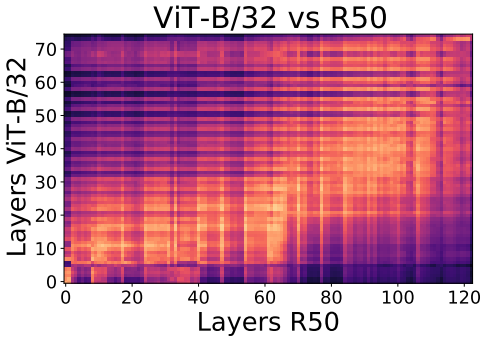
For vision transformers, we train linear probes on representations from individual tokens or on the representation averaged over all tokens, at the output of different transformer layers (each layer meaning a full transformer block including self-attention and MLP). For ResNets, we take representation at the output of each residual block (including 3 convolutional layers). The resolution of the feature maps changes throughout the model, so we perform an additional pooling step bringing the feature map to the same spatial size as in the last stage. Moreover, ResNets differ from ViTs in that the number of channels changes throughout the model, with fewer channels in the earlier layers. This smaller channel count in the earlier layers could potentially lead to worse performance of the linear probes. To compensate for this, before pooling we split the feature map into patches and flattened each patch, so as to arrive at the channel count close to the channel count in the final block. All results presented in the paper include this additional patching step; however, we have found that it brings only a very minor improvement on top of simple pooling.

B Additional Representation Structure Results

Here we include some more CKA heatmaps, which provide insights on model representation structures (compare to Figure 1, Figure 2 in the main text.) We observe similar conclusions: ViT representation structure has a more uniform similarity structure across layers, and comparing ResNet to ViT

representations show a large fraction of early ResNet layers similar to a smaller number of ViT layers.





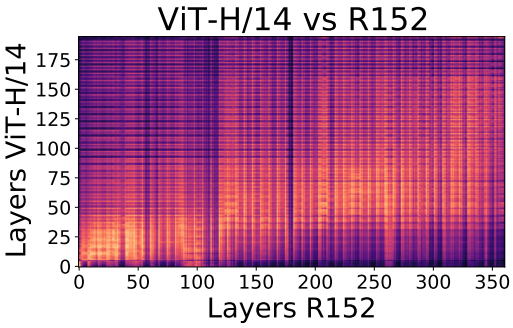


Figure B.1: Additional CKA heatmap results. Top shows CKA heatmap for ViT-B/32, where we can also observe strong similarity between lower and higher layers and the grid like, uniform representation structure. Bottom shows (i) ViT-B/32 compared to R50, where we again see that 60 of the lowest R50 layers are similar to about 25 of the lowest ViT-B/32 layers, with the remaining layers most similar to each other (ii) ViT-H/14 compared to R152, where we see the lowest 100 layers of R152 are most similar to lowest 30 layers of ViT-H/14.

C Additional Local/Global Information Results

In Figures C.1, C.2, and C.3, we provide full plots of effective receptive fields of all layers of ViT-B/32, ResNet-50, and ViT-L/16, taken after the residual connections as in Figure 6 in the text. In Figure C.4, we show receptive fields of ViT-B/32 and ResNet-50 taken *before* the residual connections. Although the pre-residual receptive fields of ViT MLP sublayers resemble the post-residual receptive fields in Figure C.1, the pre-residual receptive fields of attention sublayers have a smaller relative contribution from the corresponding input patch. These results support our findings in Section 5 regarding the global nature of attention heads, but suggest that network representations remain tied to input patch locations because of the strong contributions from skip connections, studied in Section 6. ResNet-50 pre-residual receptive fields look similar to the post-residual receptive fields.